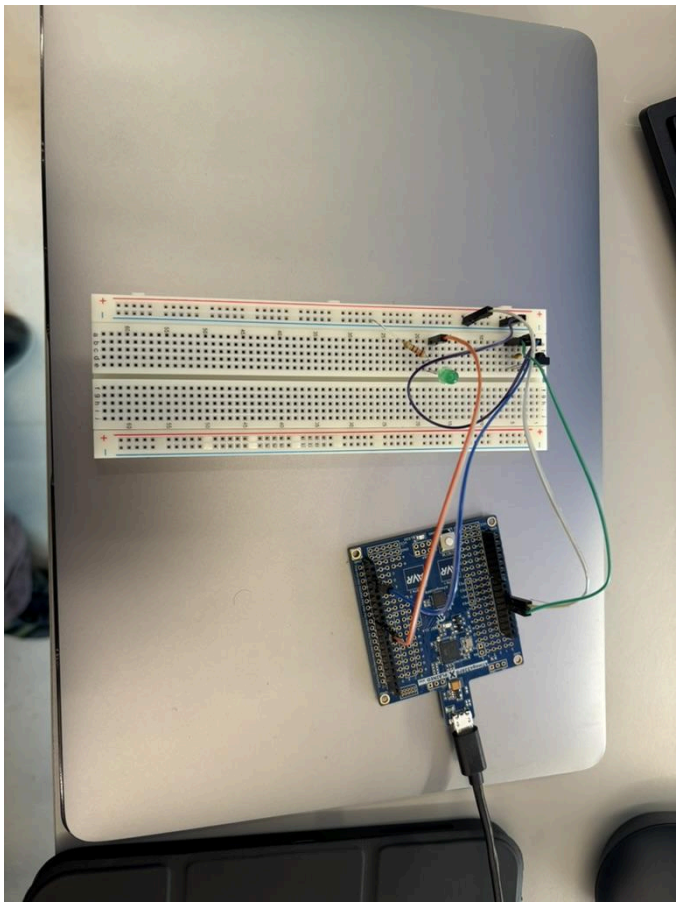
Sprint Review #1

Last week's progress

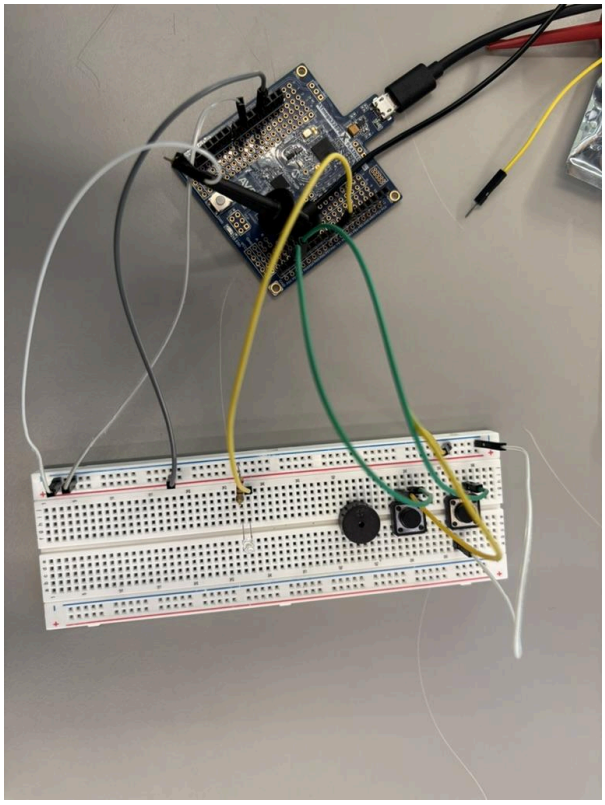
Blaster (MCU 1) — Kim & Victor:

- Trigger and reload buttons wired with GPIO and internal pull-ups, confirmed functional on breadboard.
- IR emitter circuit assembled: 940nm IR LED driven through 2N2222 NPN transistor, switching ~100mA off a GPIO-controlled base pin.
- ATmega328PB dev environment set up and running bare-metal C on the blaster MCU.



Vest (MCU 2) — Devan & Marko:

- TSOP38238 IR receiver wired with 100nF decoupling cap on VCC (required by datasheet for stable operation). Confirmed it detects 38kHz modulated IR from the blaster emitter.
- Basic hit detection prototype working: LED lights on valid IR detection via pin-change interrupt.
- Vest ATmega328PB brought up on the breadboard.



Design Change — BLE → Feather ESP32:

- Pivoted from the HM-10 BLE module to an Adafruit Feather ESP32 for the wireless link between the vest and web app. The Feather gives us Wi-Fi + BLE flexibility and a more mature software ecosystem for the web app side, while still only serving as a comms module (no application logic — all game logic remains on the ATmega328PB per course restrictions).

Current state of project

Core subsystems are individually coming online on breadboards. IR TX → RX link is proven at the physical layer. The BLE-to-Feather pivot is a scope adjustment but doesn't affect the ATmega firmware architecture — the Feather just replaces the HM-10 on the UART lines.

Hardware status: All critical components are in hand (ATmega328PBs, TSOP38238s, IR LEDs, 2N2222s, ST7735 LCD, MPU6050). Feather ESP32 is on hand. No outstanding part orders blocking progress.

Risk: IR packet reliability at 3m range hasn't been tested yet — that's a key validation target for next sprint.

Next week's plan

Task	Time	Owner	Definition of Done
Feather ESP32 + Web App BLE link	2-4 h	Marko	Feather pairs with browser via Web Bluetooth API; can send/receive test strings bidirectionally
ST7735 LCD over SPI	1-2 h	Kim	LCD displays static ammo count on blaster MCU
MPU6050 I2C bringup	1-2 h	Kim/Devan	Read raw accelerometer values over I2C at 400kHz, print to serial
Piezo buzzer tones	10 m	Victor	Timer-driven PWM produces distinct tones for shot/hit/reload events
Migrate to circuit boards (perfboard)	2 h	Victor/Devan/ Marko	Blaster and vest circuits soldered on perfboard, off breadboard